

PRATIKSHIT SINGH

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[LinkedIn](#) | [Github](#)

EDUCATION

- University of Illinois Urbana-Champaign, IL, United States** Aug 2022 - Dec 2023
Masters - Computer Science **GPA:3.9/4.0**
- Indian Institute of Technology (BHU) Varanasi, India** Jul 2016 - May 2020
Bachelor of Technology - Electrical Engineering **GPA:9.5/10.0**

TECHNICAL SKILLS

- Programming Languages:** Python, Java, TS, JS, C, C++, SQL, GraphQL, Shell, Design Patterns(LLD, HLD)
- Technologies:** Springboot, Django, Flask, Hadoop, EMR, EC2, Lightsail, PySpark, Hive, Kafka, Redis, React, Jenkins, Docker, Kubernetes, Nginx, JUnit, Mockito, Karate
- genAI:** LangChain, LangGraph, Agents, RAG, Embedding, VectorDB, Evaluation, MCP, Harness Engineering
- Databases:** Design Patterns, Indexing, Snowflake, Postgres, MySQL, DynamoDB, MongoDB, S3, Redis, HMS, Datalake
- Libraries:** PyTorch, TensorFlow, Pandas, Numpy, Sklearn, NLTK

WORK EXPERIENCE

Intuit | Software Engineer

Sep 2024 - Current

Privacy Engineering

- Building the Intuit Legal Intelligence Platform: a **LangGraph-orchestrated multi-agent system** with a **knowledge management layer**, automated request triage and routing, and end-to-end **traceability** of every agent decision via **Langfuse**, persisted to a **data lake**.
- Ideated and built Claude plugins (equipped with Skills, MCP servers, Agents, and Hooks) for Privacy Legal and Intuit-wide product teams needing privacy support, accelerating manual privacy reviews, enabling self-service triage of user queries, and **cutting privacy review turnaround from 14 days to 1 day** and **deflecting 50% of inbound privacy queries via self-service triage**.
- Spearheaded the **design & development** of **Intuit Complaint Management Platform** processing **100GBs of transcript data/day** from **datalake** with **PySpark EMR** pipelines, using **LLM models** and **RAG** to generate intelligent insights, & regulatory checks. System built with auto-evaluations, alerting, and schema-evolution support with delta tables.
- Implemented **RAG-powered** Legislative Bill Tracking application to identify upcoming changes for Intuit Privacy, accelerating legal compliance review process by **60%**.
- Spearheaded the **design & development** of multiple **event-driven, MVC** and **Hexagonal architected** SpringBoot microservices with **99.99% availability** through **multi-region, active-active setup, throttling, rate-limiting** on REST APIs.

Zeta Global | Software Engineer

June 2024 - Sep 2024

Zeta Marketing Platform

- Designed and implemented **python ETL EMR pipeline** to identify brands & category statistics in email campaigns, utilizing OCR and openAI apis to **enrich BI reporting for customers**.
- Tech Stack:** Apache Superset, Flask, OCR, DynamoDB, Boto3, openAI

Illinois Center for Transportation, UIUC | Lead Developer

Dec 2022 - May 2024

Pavement Life-Cycle Assessment Tool [[LCA application](#), [IPR application](#)]

- Led the design & development of scalable multi-stage complex web applications for Pavement Life & Cost Assessment.
- Used DRF to process requests & fetch data from Postgres, with Redux store for managing hierarchical project data in UI.
- Implemented user permissions for calculation models & tool stages with caching deployed as a multi-container application.
- Tech Stack:** Django, REST API, PostgreSQL, Redis, React, Redux, AWS LightSail, Docker Compose

JP Morgan Chase & Co | Software Engineer

Aug 2020 - Jul 2022

Quantitative Tool for Risk and PnL - Full Stack Developer

- Led the development of Athena QIT web application, created trades dashboards to view trading risk & market exposures for 15+ Global Trading Desks, using async tornado web framework and Athena Risk Engines for adhoc risk analysis.
- Designed & implemented Postgres database table & views using feeds from 200+ risk jobs. Further, implementing DB triggers for updating tables from live running risk jobs.
- Accelerated data delivery to dashboards using clustered indexing for SQL and GraphQL API calls.
- Savings** - Annual cost of \$5 Million by decommissioning legacy third-party applications.
- Stack:** Tornado-web-server, React, PostgreSQL, GraphQL, DataDog, SDLC, Automated Testing

Athena Risk Compute Batches Optimization using ML clustering algorithm - Data Scientist

- Built an EOD cron job for optimizing Risk Jobs for IR instruments using XGBoost and Linear regression models, to generate optimized batches to be processed on EC2 pods.
- Savings** - 84% reduction in total running time for Global jobs running on the Amazon EC2.
- Stack:** Feature Creation, Regression Models, Elastic-Net, AWS EC2, Cloud Computing, Kubernetes

PROJECTS

Attention Enhanced-Super Resolution GAN

[Github](#)

- Implemented SRGAN with multi-scale attention U-net discriminator & residual-in-residual dense blocks generator.
- Better- NIQE score over current SR models & accurately generated sharp edge details & textures.
- Tech Stack:** Super Resolution, GAN, Attention blocks, RRDB, U-Net, CNN

Domain Adversarial Transfer Learning for Time Series Classification

[Github](#)

- Developed methods to generate superior pre-trained 1-D CNN transfer models for time-series classification.
- **Tech Stack:** Domain adversarial networks, Error bounds, Timeseries, DTW & divergence measures, CNN

Fraud Detection on Imbalanced Class Dataset

[Github](#)

- Implemented SMOTE sampling, cross-validation methods, and cost-sensitive models on big data with 0.02% positive class.
- **Accuracy:** 97.7 % Train and 95.1% Test with cost-sensitive XGBoost.
- **Tech Stack:** EDA, Pattern mining, SMOTE, Cost-sensitive XGBoost, PCA

Deep Learning-based Fault Detection in Transmission Towers

Power Grid Corporation of India Ltd.

- Implemented CNN CV model using TensorFlow and YOLOv3 network to detect faults in various classes of Transmission Towers using high-resolution images.
- **Tech Stack:** Tensorflow, Keras, YOLOv3, CNN, sliding window, anchor box, IoU